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Received for review May 17, 1974

Accepted December 16, 1974

A Generalized Method for Predicting Second Virial Coefficients

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Expressions for predicting pure-component and cross second virial coefficients for simple and complex systems have been developed from the bound-pair formalism of Stogryn and Hirschfelder. For pure components, the generalized correlation requires the critical temperature and pressure, Thompson's mean radius of gyration or the parachor, dipole moment, and, if appropriate, a parameter to describe chemical association which depends only in the type of group (hydroxyl, amine, ester, carboxylic acid, etc.). Mixing rules have been developed for predicting cross coefficients and solvation effects can be accounted for in a similar manner to association. Agreement with experimental data on 39 nonpolar and 102 polar and associating compounds, 119 mixed nonpolar systems, and 73 mixed systems involving polar compounds, is comparable to or better than that of several other correlations including those which require data to obtain parameters. The method should be most accurate for systems of complex molecules where no data are available.

In order to accurately predict phase equilibria involving the vapor phase at pressures above atmospheric, deviations from the perfect-gas law usually need to be taken into account (Prausnitz, 1969; Nagata and Yasuda, 1974). The virial equation terminated at the second coefficient is a simple but accurate method for conditions up to a density of about one-half the critical and has been employed in completely developed methods for predicting vapor-liquid equilibria such as Prausnitz et al. (1967). Several analytical methods for predicting values for the second virial coefficient have been developed (Black, 1958; O'Connell and Prausnitz, 1967; Kreglewski, 1969; Nothnagel et al., 1973; Tsonopoulos, 1974), but except for the last, all suffer from the disadvantage of often requiring one or more parameters that must be obtained from data, or the results are too inaccurate to be acceptable.

This work develops an accurate method for predicting second virial coefficients using only critical properties and molecular parameters, all of which may usually be estimated from molecular structure to the required accuracy. From extensive comparisons with pure component and cross virial coefficient data, the present method appears to be more consistently accurate than any other purely predictive method. In addition, for strongly associating substances, the method predicts association effects at higher densities in a realistic fashion (Nothnagel et al., 1973) using a parameter which depends only on the group interaction.

Basic Expressions

The virial equation of state relates the compressibility factor to the independent intensive variables of composition, temperature, and pressure or density. Making suitable thermodynamic manipulation of this equation of

state yields the vapor phase fugacity which is used in obtaining K factors and relative volatilities. Since the accuracy of the fugacity and compressibility are about the same for the pressure-explicit and density-explicit equations truncated at the second virial coefficient (Prausnitz, 1969), and systems are usually specified by temperature, pressure, and composition, the most convenient form of the virial equation to be used is

$$z \equiv \frac{Pv}{RT} = 1 + \frac{BP}{RT} \quad (1)$$

where v is the molar volume and, in a mixture of N components

$$B = \sum_{i=1}^N \sum_{j=1}^N y_i y_j B_{ij}(T) \quad (2)$$

Here y is the mole fraction and $B_{ij}(T)$ is the second virial coefficient characterizing pair interactions between an " i " and a " j " molecule, a function only of temperature. The vapor fugacity is given by

$$f_i^v = y_i P \phi_i \quad (3)$$

where the fugacity coefficient is given by

$$\ln \phi_i = \left[2 \sum_{j=1}^N y_j B_{ij} - B \right] \frac{P}{RT} \quad (4)$$

For substances such as carboxylic acids which associate very strongly, the virial equation is not valid. However, the "chemical theory" for nonideality can give good predictions in such cases when an equilibrium constant for association is available (Nothnagel et al., 1973). Values of second virial coefficients can be related to the equilibrium constant in a simple way, so if a correlation yields accu-

rate values for such substances, it can be used for all systems.

To predict the vapor phase fugacity to 1% using eq 4 the error in the difference between second virial coefficients at 400°K should be less than about $300/P$ where the virial coefficient is in $\text{cm}^3/\text{g-mol}$ and the pressure, P , is in bars. Many systems have virial coefficient differences of the order of 100–2000 $\text{cm}^3/\text{g-mol}$ so that even at 1 bar nonideality might not be ignorable and values should be predicted to within about 100 $\text{cm}^3/\text{g-mol}$.

Values of the coefficients B_{ij} can be obtained directly from P - V - T data, from statistical mechanical formulas using an expression for the pair intermolecular potential energy, or from empirical or semitheoretical correlations. The most popular method has been the last, since the computation is usually the easiest (although not always the most accurate). In the present work statistical mechanical methods are utilized with an extended corresponding states approach to develop a predictive method for second virial coefficients. Its unique advantages are that use of a parameter which generally depends only on the "reacting" group yields good prediction for complexing systems (which is of value when no data are available) and that systems containing carboxylic acids can be correlated.

Development of Correlation

It can be shown that the various kinds of intermolecular forces contribute to the second virial coefficient in distinct ways. In particular, the contribution of nonpolar repulsion and attraction and classical electrostatic interactions can be separated rigorously from "chemical" or nonclassical interactions from charge-transfer complexing such as hydrogen bonding (Moore and O'Connell, 1971). In addition, it is possible to show that even for the classical interactions, the dynamics of pair collisions yield contributions from molecular configurations which can be described as bound, metastably bound, and free pairs (Stogryn and Hirschfelder, 1959).

$$B_{\text{total}} = B_{\text{free}} + B_{\text{metastable}} + B_{\text{bound}} \quad (5)$$

This is a particularly valuable way to consider the contributions in strongly nonideal systems since the "chemical" theory requires a bound contribution. The effective pair potential in an elastic collision is shown in Figure 1 indicating the regions of pair interaction yielding bound pairs or metastably bound pairs when the separation distance is within the well and the relative kinetic energy is lower than the maximum in the curve. It is claimed that when pairs of molecules are formed in these configurations (by collisions with a third body), they persist through several collisions with single molecules (Saran et al., 1967; Singh et al., 1967; Stogryn and Hirschfelder, 1959).

From the expressions of Stogryn and Hirschfelder, several calculations were made for the "bound" (including metastable) second virial coefficients for the Lennard-Jones and Stockmayer potentials (Saran et al., 1967; Singh et al., 1967). These numerically tabulated bound coefficients were then analytically expressed as a function of temperature and reduced dipole moment

$$B_{\text{metastable}} + B_{\text{bound}} = b_0 A \exp[\Delta H/(kT/\epsilon)] \quad (6)$$

where

$$b_0 = \frac{2\pi}{3} N_0 \sigma^3 \quad (7)$$

$$A = -0.3 - 0.05 \mu^* \quad (8)$$

$$\Delta H = 1.99 + 0.2 \mu^{*2} \quad (9)$$

$$\mu^* = \mu^2/\epsilon \sigma^3 \quad (10)$$

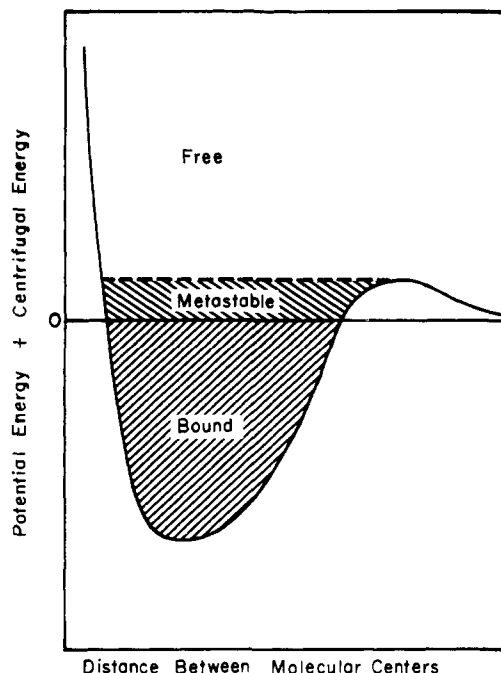


Figure 1. Physics of bound pairs.

and ϵ and σ are the effective nonpolar potential parameters (see below) while μ is the molecular dipole moment. The next sections describe the development of the correlations for B_{free} , for different classes of compounds, as well as relations to obtain ϵ and σ .

Nonpolar Substances

It is generally acknowledged (Rowlinson, 1968) that two parameter potential functions, or alternatively, two-parameter corresponding-states theory, are not adequate to correlate the properties of nonpolar substances which are different from the rare gases. However, the success of the three-parameter theories of Pitzer (1955, 1957) and others (Leland and Chapplear, 1968) indicates that only one more parameter is required to describe macroscopic properties of nonpolar substances. For polar and associating molecules, the nonpolar forces must also be determined separately in order to correctly predict the cross second virial coefficients for these substances and nonpolar molecules (Rigby et al., 1969). Use of the common third parameters of acentric factor or critical compressibility factor should not be employed for this because they are affected by the polarity and complexing interactions as well.

Although the homomorph concept of Bondi and Simkin (1956, 1957) has been used previously (O'Connell and Prausnitz, 1967) to obtain effective acentric factors as the third parameter, it was felt desirable here to seek a molecular structure parameter to describe nonsphericity of the nonpolar forces. The mean radius of gyration used by Thompson and Braun (1968) appeared to be the most accessible and thus was adopted. For linear molecules it is defined as

$$R' = \sqrt{(I_A I_B)^{1/2}/m} \quad (11)$$

while for nonlinear molecules the definition is

$$R' = \sqrt{2\pi(I_A I_B I_C)^{1/3}/m} \quad (12)$$

where the I s are principal moments of inertia and m is the molecular mass. In the present correlation, the quantities are chosen so that the units of R' are Å.

The method for including the nonspherical nonpolar forces was to obtain an effective nonpolar acentric factor,

ω' from the mean radius of gyration and to assume that only the free contribution of the second virial coefficient would be affected by the nonsphericity. This is based on the assumption that it is the repulsive portion rather than the attractive portion of the potential which is mostly affected by nonpolar nonsphericity (Rowlinson, 1968). From some available data, primarily on hydrocarbons, an analytic expression for nonpolar substances was developed

$$B_{\text{free-nonpolar}} = b_0(0.94 - 1.47/T^{*'} - 0.85/T^{*'}{}^2 + 1.015/T^{*'}{}^3) \quad (13)$$

where

$$1/T^{*'} = \epsilon/kT - 1.6\omega' \quad (14)$$

and

$$\omega' = 0.006R' + 0.02087R'^2 - 0.00136R'^3 \quad (15)$$

where R' is expressed in Å. The values for ϵ and σ for nonpolar molecules are obtained from a correlation similar to that of Tee et al. (1966)

$$\epsilon/kT_c = (0.748 + 0.91\omega') \quad (16)$$

and

$$\sigma = (2.44 - \omega')(T_c/p_c)^{1/3} \quad (17)$$

where T_c is in degrees Kelvin and p_c in atmospheres.

Polar, Nonassociating Substances

Although the mean radius of gyration provides a strictly "nonpolar" third parameter, for polar substances with large dipole moments (here $\mu > 1.45$) the critical properties are undoubtedly affected by polarity. In order to compensate for this in the process of obtaining values for the nonpolar parameters ϵ and σ , the device of angle averaging was employed to determine the effect of polarity (Cook and Rowlinson, 1953). The n -6 potential with an additional dipole term can be written

$$\Gamma_{\text{total}} = C\epsilon \left[\left[\frac{\sigma}{r} \right]^n - \left[\frac{\sigma}{r} \right]^6 \right] + \frac{\mu^2}{r^3} g(\Omega) \quad (18)$$

where

$$C = n^{n/n-6}/(n-6)6^{6/n-6} \quad (19)$$

and $g(\Omega)$ is an orientation factor. Using the free-energy averaging procedure, Γ_{total} can be approximated to the first order as

$$\Gamma_{\text{total}} = C\epsilon' \left[\left[\frac{\sigma'}{r} \right]^n - \left[\frac{\sigma'}{r} \right]^6 \right] \quad (20)$$

$$\epsilon = \epsilon'/[1 + \xi]^{n/n-6} \approx$$

$$\epsilon' \left\{ 1 - \left(\frac{n}{n-6} \right) \xi \left[1 - \left(\frac{n}{n-6} + 1 \right) \xi/2 \right] \right\} \quad (21)$$

$$\sigma^3 = \sigma'^3 [1 + \xi]^{3/(n-6)} \approx \sigma'^3 [1 + 3\xi/(n-6)] \quad (22)$$

$$\begin{aligned} \xi &= 4\mu^4/3C\epsilon'\sigma'^6kT_c \\ &= k\mu^4/(5.723 \times 10^{-8}C\epsilon'\sigma'^6T_c) \end{aligned} \quad (23)$$

where p_c is in atm, T_c in °K, and μ in debyes (10^{-18} esu). The values of ϵ' and σ' are obtained from eq 16 (or 30, see below) and 17, respectively. A rough correlation for n (obtained by fitting normal paraffin second virial coefficient data, Vives, 1971), is

$$n = 16 + 400\omega' \quad (24)$$

and a good correlation for C is

$$C = 2.882 - 1.882\omega'/(0.03 + \omega') \quad (25)$$

the values of ϵ and σ obtained above are then used in eq 7, 10, and 14 for calculating the nonpolar portions of the sec-

ond virial coefficient. The forms of eq 21, 22, and 25 were developed to minimize computation time which could be excessive if the exact expressions were used.

The polar contributions to B_{free} were correlated empirically by using data on some halogenated and oxygenated substances, which do not associate, and SO_2 . The final expression is

$$B_{\text{free}} = B_{\text{free-nonpolar}} - b_0\mu^{*'}(0.75 - 3/T^{*'} + 2.1/T^{*'}{}^2 + 2.1/T^{*'}{}^3) \quad (26)$$

where

$$\begin{aligned} \mu^{*'} &= \mu^* - 0.25 & \mu^* &\geq 0.25 \\ &= 0 & 0.25 > \mu^* &\geq 0.04 \\ &= \mu^* & 0.04 > \mu^* &\geq 0 \end{aligned} \quad (27)$$

Polar, Associating Substances

With the above expressions developed, examination was made of the data for water and alcohols, esters, amines, mercaptans, ketones, and all others where the predicted values were not as negative as the data and the possibility of association could exist. Adopting the idea that the association contribution was separate (Moore and O'Connell, 1971) and could be correlated as an equilibrium constant, the second virial coefficient for the substances is

$$B_{\text{total}} = B_{\text{free}} + B_{\text{metastable}} + B_{\text{bound}} + B_{\text{chem}} \quad (28)$$

where for the present work

$$B_{\text{chem}} = b_0 \exp\{\eta[650/(\epsilon/k + 300) - 4.27]\} \times \{1 - \exp[1500\eta/T]\} \quad (29)$$

The particular form of eq 29 was adopted to conform to the limit $B_{\text{chem}} = 0$ when $T = \infty$, to use only one parameter for each group, but recognize some deviations within members of the group as indicated by data for hydroxyls, amines, ketones, and esters, and to note that there is usually a correlation between the enthalpy and entropy of reaction (association) (Cheh et al., 1966). Association should affect the critical properties, so the value of ϵ' for use in eq 21 was modified to be

$$(\epsilon'/kT_c) = 0.748 + 0.91\omega' - 0.4\eta/(2 + 20\omega') \quad (30)$$

This is likely to be the weakest link in the correlation and we feel that if any parameter is to be fitted to data in further application of the correlation, the best improvement over the generalized method would be to fit ϵ/k . We have not done this here, since our emphasis is on the generalized prediction.

In the "chemical" theory used for carboxylic acids, the equilibrium constant for the formalism described by Nothnagel et al. (1973) is obtained from

$$K_p = - \frac{B_{\text{bound}} + B_{\text{metastable}} + B_{\text{chem}}}{RT} \quad (31)$$

and their molecular volumes are replaced by B_{free} in their expressions for vapor-phase nonideality. For organic acids, the results of eq 31 are better than any previously obtained when the value of η is 4.5 and eq 29 has the factor $650/(\epsilon/k + 300)$ replaced by $42,800/(22,400 + \epsilon/k)$. It is recommended that eq 31 be used only for systems containing carboxylic acids, however, since the values of B_{total} of all other substances were evaluated from fitting the data by the virial equation of state, not from the "chemical" theory. It has not been proven one way or another, but in cases where B is very negative, use of the chemical theory could provide better values at higher densities and in complex systems than does eq 4 (Nagata and Yasuda, 1974).

Table I. RMS Deviations for Pure Component Virial Coefficients from Several Correlations

Compound	No. of data	Temperature range, °K	RMS deviation, cm ³ /g-mol					
			Hayden	Extended Pitzer	Nothnagel et al.	Tsonopoulos	Black	Kreglewski
Xenon	7	275–598	6.0	1.4	7.6 ^b	1.5	2.8	3.1
Sulfur dioxide	23	265–473	10.0	59.7	16.1 ^b	55.8	8.8	95.1
Cyanogen	6	308–423	42.4	59.4	15.9 ^b	53.8	46.2	...
Methane	16	110–600	4.6	3.8	13.9 ^b	4.0	24.9	2.1
Propane	13	260–550	4.2	12.2	4.1 ^b	8.5	4.0	5.6
<i>n</i> -Heptane	16	353–598	19.7	36.7	53.2 ^{a,b}	13.2	175.1	13.8
Benzene	16	337–473	36.7	36.9	36.4 ^{a,b}	32.7	47.7	43.0
<i>cis</i> -2-Butane	5	260–343	52.5	58.2	18.6 ^b	65.4	54.5	...
Tetrachloromethane	5	340–420	96.6	121.1	96.7 ^b	122.4	162.0	74.9
Chloromethane	12	280–623	34.5	29.4	17.1 ^b	45.8	19.0	39.0
Water	9	373–578	31.7	...	...	...	...	131.2
			25.8 ^b	64.9 ^b	7.6 ^b	80.2 ^{a,b}	9.8	...
1-Propanol	4	378–473	19.1	...	...	...	...	247.0
			10.6 ^b	25.8 ^b	10.1 ^b	20.2 ^b	...	...
Isobutyl alcohol	4	393–439	66.6	...	...	...	...	245.6
			55.2 ^b	108.3 ^b	31.6 ^b	26.8 ^b	...	...
Ethoxy ethane	10	298–400	67.7	36.7 ^b	27.4 ^b	57.3	50.9	104.5
2-Butanone	4	339–371	31.7	...	...	57.6	...	572.0
			31.7 ^b	44.3 ^b	77.5 ^b	...	...	...
Furan	3	280–305	9.7 ^b	18.1 ^b	16.1 ^b	...	...	...
Acetic acid	10	355–445	4.1(%)	...	54.3(%)	...	...	...
Methylamine	11	293–405	26.2	...	...	32.7	...	123.4
			23.4 ^b	31.9 ^b	26.5 ^b	...	...	...
Pyridine	7	366–438	132.1	...	...	274.3	...	49.8
			76.9 ^b	40.8 ^b	36.8 ^b	...	...	...
<i>n</i> -Thiopropene	2	321–341	52.8	...	...	179.4	...	...
			10.7 ^b	35.8 ^b	29.6 ^b	...	...	...

^a Apparently present data different from that used to obtain empirical parameters. ^b One parameter fitted to second virial coefficient data. Otherwise generalized method of group association parameter used.

Cross Coefficients

Since the above expressions have a molecular basis, except for adopting mixing rules for the parameters, the method of calculating cross coefficients must be consistent with them. On the basis of extensive comparisons for systems with two nonpolar and two polar substances, the following rules were chosen

$$\epsilon_{ij} = 0.7(\epsilon_i \epsilon_j)^{1/2} + 0.6(1/\epsilon_i + 1/\epsilon_j) \quad (32)$$

$$\sigma_{ij} = (\sigma_i \sigma_j)^{1/2} \quad (33)$$

$$\omega_{ij}' = 0.5(\omega_i' + \omega_j') \quad (34)$$

$$\mu_{ij}^* = \mu_i \mu_j / \epsilon_{ij} (\sigma_{ij})^3 \quad (35)$$

and $\eta_{ij} = 0$ unless the species are in the same group or a special solvation contribution could be justified and empirically determined for each pair of groups. Here the σ_i and ϵ_i are calculated from eq 21 and 22, respectively.

In addition, to account for the effect of induction for polar (*i*) – nonpolar (*j*) systems when $\mu_i > 2.0$ angle averaging was again employed to give values of ϵ_{ij}^{pn} and σ_{ij}^{pn} to be used in eq 7, 10, 14, 25, and 28 to obtain B_{ij} (*n* is calculated from eq 24 and 34)

$$\epsilon_{ij}^{pn} = \epsilon_{ij} [1 + \xi^{pn} n / (n - 6)] \quad (36)$$

$$(\sigma_{ij}^{pn})^3 = (\sigma_{ij})^3 [1 - 3\xi^{pn} / (n - 6)] \quad (37)$$

$$\xi^{pn} = \mu_i^2 (\epsilon_j)^{2/3} (\sigma_j)^4 / [\epsilon_{ij} (\sigma_{ij})^6] \quad (38)$$

where ϵ_{ij} and σ_{ij} are found using eq 32 and 33.

The form of eq 38 was chosen from assuming that the average polarizability of the molecules varies as $\epsilon^{2/3} \sigma^4$ as implied by equating the dispersion attraction coefficient in the r^{-6} term in eq 18 to the Slater-Kirkwood formula (Hirschfelder et al., 1964) ($\epsilon \sigma^6$ varies with $\alpha^{3/2}$). The factor of unity in eq 38 was adopted on the basis of the available data. Again, the weakest link in the general correlation is probably eq 32 and when data are available, they could be used to fit a value of ϵ_{ij} for use at other temperatures.

Comparisons with Data and Other Correlations

Complete tables of parameters and comparisons are presented in supplementary material. For illustrative purposes, Table I shows, for representative pure component systems, RMS deviations for second virial coefficients using eq 6–10, 17, 21–30, and those obtained by using the methods of O'Connell and Prausnitz (1967) (so-called Extended Pitzer method), Black (1958), Kreglewski (1969), Nothnagel et al., (1973), and Tsonopoulos (1974). Except for carboxylic acids, data for virial coefficients at temperatures where the saturation pressure was less than about 500 mm Hg and where the temperature was above 600°K were systematically excluded from the comparisons since errors in calculation of deviations from ideal gas behavior are small under these conditions and the experiments are subject to large uncertainties. Many of the data were taken from Dymond and Smith's Critical Compilation (1969), although systems were generally ignored if all of the generalized correlations had deviations significantly

Table II. Comparison of Second Virial Coefficient Correlations for Classes of Pure Compounds

Class	No. of sub- stances	No. of data	Hayden	Average substance RMS deviation, ^a cm ³ /g-mol				
				Extended Pitzer	Nothnagel et al.	Tsonopoulos	Black	Kreglewski
Monatomics, diatomics,	21	259	20.8(0)	24.4(1)	16.1 ^b (1)	18.2(0)	20.8(0)	27.0(0)
triatomics, tetraatomics, SF ₆	17 ^c	221	11.6(0)	13.7(0)	9.7 ^b (0)	11.7(0)	15.1(0)	23.0(0)
Paraffins	14	167	27.8(0)	25.0(0)	32.8 ^b (0)	19.6(0)	100.8(6)	26.2(1)
Cycloparaffins, aromatics	7	53	84.7(4)	70.9(1)	46.4 ^b (0)	72.2(1)	145.6(6)	115.9(4)
Olefins, acetylenes	14	405	62.6(4)	57.4 ^d (3)	28.3 ^b (1)	53.1 ^d (1)	98.5 ^d (7)	...
Halogenated	29	261	50.4(3)	80.1(10)	30.2 ^b (0)	74.3(7)	...	...
	26 ^e	247	40.7(0)	72.4(9)	32.4 ^b (0)	52.2(5)	...	...
Alcohols, water	10	81	60.7(2)	...	...	...	...	283.3(9)
	10	81	44.8 ^b (1)	66.2 ^b (1)	56.6 ^b (1)	54.4 ^b (1)	...	...
	9 ^f	58	43.8 ^b (1)	59.6 ^b (1)	51.3 ^b (1)	34.0 ^b (0)	...	...
Other oxygenated	18	123	72.8(3)	...	...	79.7(4)	...	...
	18	123	58.8 ^b (2)	45.5 ^b (1)	...	...	...	...
	12 ^g	106	47.0(0)	...	...	42.8(0)	...	310.6(10)
	12 ^g	106	42.0 ^b (0)	48.5 ^b	58.2 ^b (1)	...	...	...
Organic acids	5	55	4.5(%)	...	48.6 ^h (%)	...	...	...
Nitrogen substituted,	13	97	56.5(2)	...	...	94.9(5)	...	...
ammonia	13	97	33.8 ^b (0)	59.7 ^b (2)	63.8 ^b (2)	...	...	...
Sulfur substituted	15	43	95.2(3)	...	...	146.2(9)	...	...
	15	43	72.4 ^b (0)	58.6 ^b (0)	67.46(0)	...	...	...
All	141	1189	49.6(21)	...	...	62.8(28)	...	...
	141	1189	40.7 ^b (14)	53.2 ^b (19)	35.2 ^b (5)	...	...	...
	127	1097	43.5(14)	...	...	52.2(21)	...	...
	127	1097	39.0 ^b (9)	50.5 ^b (17)	32.5 ^b (5)	...	...	...

^a Average of RMS deviations of all substances in class weighted by unity for number of data being from 1-9, by two for 10-19 data, by three for 20-29 data, etc. Number of systems with RMS deviation greater than 100 cm³/g-mol shown in parentheses. ^b Calculations using one parameter determined from second virial coefficient data on individual substance. Otherwise completely generalized method or single association parameter used for all substances in group (hydroxyl, ketone, formate, other ester, aldehyde, carboxylic acid, amine, mercaptan, nitro, cyano). ^c Without carbon disulfide, fluorine, hydrogen cyanide, cyanogen. ^d Polar substances treated with zero dipole moment to reduce errors. ^e Without bromoethane, chloropropane, *t*-dichloroethene. ^f Without methanol. ^g Without methoxy ethane, 1-ethoxypropane, 1-methoxypentane, 2-pentanone, 3-pentanone, methoxybenzene. ^h Without heptanoic acid (572.4% RMS dev.).

greater than 100 cm³/g-mol when they correlated other similar systems adequately. In Table II all of the substances considered have been broken down into classes and (weighted) average RMS deviations for the substances in each class are shown in several ways. First, some of the classes have substances for which the data are scattered or perhaps suspect. In these cases, the results are listed for all the substances, and then for the systems where the agreement is more satisfactory for all the correlations. Generally, the deviations of all of the correlations are reduced by 10-20 cm³/g-mol by removing only a few substances from consideration where deviations were several hundred cm³/g-mol on occasion. Another discrimination made is between those correlations which fit virial coefficient data such as that of Nothnagel et al. (1972) and those which predict values using parameters from other information, such as the present one and that of Tsonopoulos. The Extended Pitzer method used empirical parameters only for complex molecules so is generalized for some classes and empirical for others. In some cases, new association factors were determined for the Extended Pitzer correlation. For the method of Tsonopoulos, his generalized correlation for polar effects was used for all substances except those with hydroxyl groups. The comparisons here for Black (1958) and Nothnagel et al. (1973) are occasionally unfair because sometimes they obtained their parameters by fitting different data than those used here. However, this should not affect the average deviations by more than a few cm³/g-mol.

Table III shows RMS deviations from experiment of the correlations for representative unlike systems. Table IV

shows all of these systems broken down into classes. As suggested by Prausnitz (1969), for the Extended Pitzer Correlation, the critical temperature for unlike interactions was calculated from the equation

$$T_{c_{ij}} = (T_{c_i} T_{c_j})^{1/2} v_{c_i} v_{c_j} [(v_{c_i}^{1/3} + v_{c_j}^{1/3})/2]^6 \quad (39)$$

This made significant improvement in this correlation when the substances differed in size. For the correlations of Tsonopoulos (1974), eq 39 was also used for all nonpolar systems. However, individually fitted deviations from the geometric mean were used for essentially all other classes of systems. The prescriptions for cross coefficients given by Black (1958), Kreglewski (1969), and Nothnagel et al. (1973) were used for their correlations.

Discussion

As the results of Tables I through IV show, none of the correlations is consistently "best" and all can yield large errors (>100 cm³/g-mol) for several systems. It is not clear what the experimental uncertainties are but they can be large (Dymond and Smith, 1969, Nothnagel et al., 1973), and we have not attempted as critical an evaluation as Tsonopoulos (1974) made. Some evidence of error is when the fitted parameters of Nothnagel et al. do not follow an orderly progression through a homologous series of compounds.

To determine whether one method is "best" of all considered requires evaluation of the job to be done and the information available. We believe correlations are essen-

Table III. Comparison of Correlations with Experimental Data for Some Cross Second Virial Coefficient Systems

System	No. data	Temperature range, °K	RMS deviations, cm ³ /g-mol					Kreglewski
			Present	Extended Pitzer ^a	Nothnagel et al.	Tsonopoulos ^a	Black	
Argon-nitrogen	9	90-323	5.5	5.8	15.6	7.5	20.6	5.4
Hydrogen- <i>n</i> -octane	6	473-573	10.5	27.2	66.7	16.5	30.6	10.5
Methane-neopentane	8	303-403	8.4	5.3	30.9	3.6	18.7	24.8
Methane- <i>n</i> -pentane	13	298-511	15.5	23.6	14.1	22.6	17.0	16.1
Methane-naphthalene	6	294-341	15.0	18.9	...	23.3	248.3	94.2
<i>n</i> -Heptane-benzene	5	463-503	12.8	13.1	27.6	21.6	41.7	19.2
Propane-methyl bromide	13	218-321	43.7	52.0	35.7	84.6	...	70.8
Carbon tetrachloride-chloroform	6	310-343	68.4	123.9	98.9	73.6	143.7	48.7
<i>n</i> -Butane-acetone	4	282-321	48.4	247.6	38.5	43.9*	226.8	159.7
Ethyl chloride- <i>n</i> -propyl chloride	3	303-323	93.2	189.6	90.4	244.7	...	...
Methyl iodide-diethyl ether	4	313-358	107.8	160.6	124.2	267.0	...	119.2
Methyl chloride-acetone	6	323-428	17.9 ^b	54.7 ^b	38.3 ^b	13.1 ^b	7.1	164.7
Chloroform-ethyl acetate	4	323-368	123.2 ^c	63.0 ^b	65.9 ^b	56.1 ^b	...	...
Chloroform-propyl formate	4	324-368	39.8 ^c	42.9 ^b	78.9 ^b	49.2 ^b	...	...

^a Modification of geometric mean for T_c according to Prausnitz (1969), p 130. $T_{c_{ij}} = (T_{c_i} T_{c_j})^{1/2} [(64 v_{c_i}/v_{c_j})/(1 + (v_{c_i}/v_{c_j})^{1/3})^6]$ except where empirical value given. * ^b Individual system solvation parameter. ^c Group solvation parameter.

Table IV. Comparisons of Several Correlations for Cross Second Virial Coefficients for Classes of Systems

Class	No. of systems	No. of data	Average system RMS deviation, cm ³ /g-mol ^a					Kreglewski
			Hayden	Extended Pitzer	Nothnagel et al.	Tsonopoulos	Black	
Monatomic, nonpolar	64 ^b	274	10.3(0)	13.5(2)	24.5(2)	13.7(2)	32.8(4)	25.7(2)
Diatomic								
Triatomic-nonpolar	76	316	16.7(3)	17.8(3)	...	17.2(3)	60.1(13)	43.6(7)
Other nonpolar-nonpolar (except fluorocarbons)	31 ^b	167	18.4(1)	21.8(1)	20.9(0)	23.1(1)	23.6(1)	20.5(1)
	34	187	20.7(1)	24.2(1)	...	24.5(1)	56.1(4)	34.6(3)
Hydrocarbon-fluorocarbon	9	24	15.4 ^c (0)	11.5 ^d (0)	35.3 ^d (0)	13.4 ^f (0)	...	...
Small nonpolar (to C ₃ H ₈)-polar (nonsolvating)	24 ^c	95	29.5(1)	49.7(2)	41.0(1)	29.0 ^d (1)	72.7(2)	55.8(1)
	31	111	28.0(1)	46.9(2)	41.6(1)	31.9 ^d (1)	...	56.2(1)
Large nonpolar-polar (nonsolvating)	13	46	60.6(2)	189.6(9)	109.6(6)	116.4(6)	232.7(11)	149.2(8)
Polar-polar (nonsolvating)	5	16	132.3(3)	113.8(2)	144.2(4)	187.7(3)	...	76.5(1)
Solvating	24	91	51.5 ^d (6)	85.3 ^d (4)	60.6 ^d (3)	60.0 ^d (2)	...	...
All	177	729	26.6(13)	43.2(18)	40.7(15)	34.8(15)	...	...

^a Weighted average RMS deviations for systems in class with weight of unity for 1-5 data, of two for 6-10 data, of three for 11-15 data, etc. Number of systems with RMS deviation greater than 100 cm³/g-mol in parentheses. ^b Does not include: systems with *i*-C₈H₁₈, *n*-C₁₀H₂₂, *n*-C₁₆H₃₄, *n*-C₁₈H₃₈, naphthalene. ^c Does not include: systems with hydrogen sulfide, boron trifluoride, bromomethane, 2-propanol, 2-butanone, diethylamine, nitromethane. ^d Individual system parameter used for many or all systems; otherwise, group parameter or generalized method used. ^e Solvation parameter calculated from $\eta = 3 \times 10^{-5} \epsilon/k - 7 \times 10^{-7} (\epsilon/k)^2 - 3 \times 10^{-23} (\epsilon/k)^8$ (< 0). ^f $T_{c12} = 0.9(T_{c1} T_{c2})^{1/2}$.

tially equivalent if their average deviations are within 5-10 cm³/g-mol of each other. Thus, for the pure component categories of small molecules all the correlations are about the same. For the larger nonpolar substances the correlations of Black and Kreglewski fail badly while the other generalized correlations are not too different from each other and from the empirical correlations. For the oxygenated polar and complexing substances the present generalized method and that of Tsonopoulos are similar, but the present correlation is significantly better in systems involving halogens, nitrogen, and sulfur atoms. Part of this is due to the fact that Tsonopoulos' generalized method was used, though he preferred a different form for each class of compounds. In these cases the empirical correlations appear somewhat better than the generalized ones, as they do not yield large errors (>100 cm³/g-mol)

for as many systems. However, this may be a problem of inaccurate data rather than failure of the generalized correlations.

For cross coefficients, all of the methods yield similar results although the present one is marginally "best" in nearly all cases.

For the larger nonpolar substances with polar substances, the present one is the only one that is acceptable. In the few polar-polar mixtures available, the data are apparently of poor quality since only the correlation of Kreglewski does adequately.

In general, the present method seems to be the most reliable generalized method and offers a good framework for empirical improvements, but only by 10-20 cm³/g-mol on the average over the method of Tsonopoulos. This is really of marginal significance. On the other hand, its edge in

accuracy in both pure components and unlike systems as well as its unique feature of predicting organic acid association to high precision may make it worthwhile to implement for all systems in design methods for moderate pressure. Another advantage is that in systems containing mixtures of permanent gases and condensed substances such as paraffins with carbon number greater than 8 and solids like naphthalene, predictions can be made even though there are no virial coefficient data to obtain empirical pure component parameters for the heavy substances. Calculations of nonideality are important here because vapor phase nonideality significantly affects the concentration of the heavy component in the vapor (Prausnitz, 1969).

Although the values of η could be interpreted as enthalpies of association and the ϵ and σ of eq 21 and 22 may be close to the "true" nonpolar parameters of the substances, we have placed no great significance on them because of the constraints put on the development of the correlation. On the other hand, the relative success of the approach may warrant further consideration about their meaning.

It is not clear whether such substances as acetonitrile and nitromethane do actually associate. Another explanation for the need to use an association constant is that for these two highly polar substances ($\mu^* = 3.6$ and 2.7 , respectively, whereas all other substances considered except water had values of μ^* less than 2) the form of the correlation is inadequate. However, the prediction of cross coefficients involving these species is quite satisfactory as is. The present method avoids association factors for chloroform and other halogenated paraffins and for ethers and sulfides, where there is no evidence of association, but which required parameters for satisfactory prediction in the correlation of O'Connell and Prausnitz (1967).

To obtain the parameter R' when it is not available in the complete tabulations, in Thompson and Braun (1968), or in the thesis of Thompson (1969), it must be calculated. A computer program has been included with the data tabulations for computation of the values when the Cartesian coordinates of the atoms and their masses are supplied. These are available in the tabulations of Sutton (1958, 1965). A satisfactory alternative method of calculating R is from the parachor as described by Harlacher and Braun (1970). While the correlation is not as good as implied by the few systems shown by Harlacher and Braun, R' for eq 15 can be obtained to within ± 0.3 from the equation

$$P' = 50 + 7.6R' + 13.75R'^2 \quad (40)$$

A deviation of 0.3 in R' gives differences of the order of $100 \text{ cm}^3/\text{g-mol}$ in second virial coefficients for the substances which eq 40 is poorest. This accuracy may be good enough for many purposes. Values of μ can be estimated to fair accuracy by the bond addition method of Smyth (1955). For new groups not used in this study, values of η must be determined empirically. In addition, for those groups and cross interactions where only one system has been studied the values should be subject to reevaluation as more data appear.

Conclusions

A successful correlation for predicting both pure component and cross second virial coefficients has been developed using molecular concepts together with empirical modifications. Requiring only critical temperature and pressure, dipole moment, mean radius of gyration or parachor, and a "chemical" interaction parameter which depends only on the associating group (but can be fitted to data) the accuracy of this predictive correlation is ade-

quate for calculating vapor-liquid equilibria up to moderate pressures. The correlation is generally as good as any other available method for simple substances and is often significantly more accurate for complex systems. For systems where no data are available, this method appears to offer the most reliable completely predictive framework.

Acknowledgment

The authors are grateful to Thomas Duncan for computational assistance, to J. M. Prausnitz for helpful correspondence, and to the NorthEast Regional Data Center of Florida for use of its facilities.

Nomenclature

- A = parameter in present correlation, from eq 8
- B = second virial coefficient, $\text{cm}^3/\text{g-mol}$
- b_0 = equivalent hard-sphere volume of molecules, $\text{cm}^3/\text{g-mol}$, from eq 7
- C = constant in n -6 potential, from eq 19 or 25
- f_i^v = vapor phase fugacity, atm
- ΔH = effective enthalpy of formation of physically bound pairs, ergs/molecule, from eq 9
- I = molecular moment of inertia, $\text{g } \text{\AA}^2$
- k = Boltzmann constant = 1.3805×10^{-16} , ergs/molecule $^\circ\text{K}$
- K_p = equilibrium constant for "chemical" theory of vapor nonideality, atm^{-1} , eq 31
- m = molecular mass, g
- n = exponent parameter in n -6 potential model, from eq 24
- N_0 = Avogadro's number = 6.0225×10^{23} molecules/mol
- P = pressure, atm
- P' = parachor, used in eq 40
- r = separation of molecular centers of mass, $\text{\AA}$
- R = universal gas constant = $82.054 \text{ cm}^3 \text{ atm/g-mol } ^\circ\text{K}$
- R' = mean radius of gyration, $\text{\AA}$, from eq 11, 12, or 40
- T = absolute temperature, $^\circ\text{K}$
- T^* = reduced temperature, eq 14
- v = molar volume, $\text{cm}^3/\text{g-mol}$
- y_i = vapor mole fraction of species i
- z = compressibility factor, eq 1

Greek Letters

- Γ_{total} = molecular pair potential energy, ergs/molecule, eq 18 and 20
- ϵ = energy parameter for use in eq 6, 10, and 14, ergs/molecule for pure nonpolar pairs, from eq 16; for pure polar and associating pairs, from eq 21; for unlike nonpolar pairs and polar pairs from (31); for unlike polar-nonpolar pairs, from eq 35
- ϵ' = energy parameter for pure polar and associating pairs for use in eq 21 and 23, ergs/molecule, from eq 30
- ξ = angle averaged polar effect for pure substances, from eq 23
- ξ^{pn} = angle averaged polar effect for unlike polar-nonpolar interactions, from eq 38
- η = association parameter for pure interactions, solvation parameter for unlike interactions
- μ = molecular dipole moment, D (10^{-18} esu)
- μ^* = reduced dipole moment for use in eq 8, 9, 27, from eq 9 for pure interactions, eq 34 for unlike interactions
- $\mu^{*'} = \text{reduced dipole moment for use in eq 26, from eq 27}$
- σ = molecular size parameter $\text{\AA}$, for use in eq 7 and 10. For pure nonpolar pairs, from eq 17; for pure polar and associating pairs from eq 22, for unlike nonpolar pairs and polar pairs from eq 32; for unlike polar-nonpolar pairs from eq 36
- σ' = molecular size parameter for pure polar and associating pairs for use in eq 22, $\text{\AA}$, from eq 17
- ϕ_i = vapor phase fugacity coefficient, from eq 4
- ω' = "nonpolar" acentric factor for use in eq 14, 16, 17, 24, 25, 30, from eq 15 for pure interactions and eq 33 for unlike interactions
- Ω = orientation angles for dipolar interactions

Subscripts

C = critical property

ij = from interaction of species i with species j

bound = for physically bound pairs of molecules, eq 6

chem = for chemically bound pairs of molecules, eq 29

free = for unbound pairs of molecules, eq 26

metastable = for metastably bound pairs of molecules, eq 6

total = for all pair interactions, eq 28

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Received for review October 5, 1973

Accepted March 11, 1975

Supplementary Material Available. Complete tables of parameters and comparisons will appear following these pages in the microfilm edition of this volume of the journal. Photocopies of the supplementary material from this paper only or microfiche (105 × 148 mm, 24× reduction, negatives) containing all of the supplementary material for the papers in this issue may be obtained from the Journals Department, American Chemical Society, 1155 16th St., N.W., Washington, D.C. 20036. Remit check or money order for \$4.50 for photocopy or \$2.50 for microfiche, referring to code number PROC-75-209.

An Optimization Study of the Pyrolysis of Ethane in a Tubular Reactor

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Optimum temperature profiles during the pyrolysis of ethane exist because the yield goes up with increasing temperature, but consequently, the reactor must be shut down and cleaned out with increasing frequency because the carbon formed deposits along the reactor wall causing high pressure drop. The combined effect causes the yearly production of ethylene to go through an optimum. To find this optimum, a computer program was developed with the ability of handling 25 simultaneous reactions involving up to 25 components. It calculates the carbon deposition profile and the changing pressure profiles, as a function of a predetermined reaction gas temperature profile. The reactor will remain in production until the inlet pressure exceeds 8 atm. The average yearly production rate is calculated, assessing a reactor shut down penalty of 24 and 48 hr required for the cleaning of the clogged pyrolysis tubes. The optimum exit temperature for the 24-hr penalty was 1127°K with a corresponding 59% one pass ethane conversion. The 48-hr penalty lowers the optimum exit temperature to 1124°K and a 50.5% ethane conversion. The practice of increasing pressure to compensate for carbon buildup results in accelerated carbon deposition and is detrimental to the overall production scheme.

Introduction

To perform an optimization one needs some sort of plant description to form an objective function such as production rate or profit margin which must be optimized in terms of the independent variables.

Historically, plant data were used in deriving mathematical models by regression analysis. Some plants had even been deliberately disturbed in order to obtain enough

data to determine the independent variables into which the plant was being fitted (Shah, 1967).

This method has many drawbacks such as noise in the plant data causing unreliability in the readings, and a limited range of conditions under which the data are collected. Conditions outside of the range of those specifically studied must be calculated by the relatively unreliable method of extrapolation.